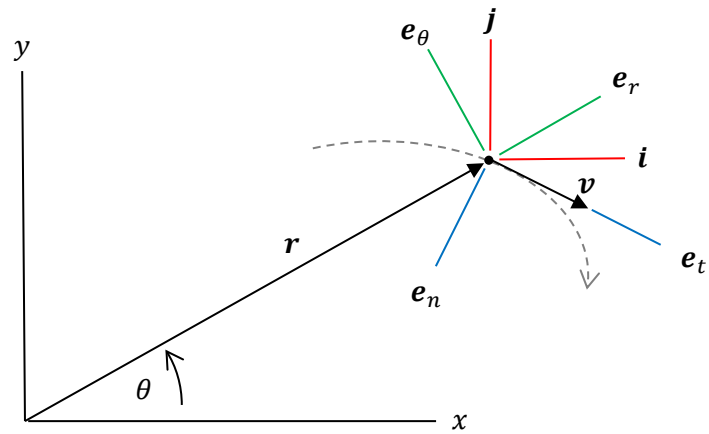
Dynamics "Vocabulary"

- Expressions of position, velocity, and acceleration in 2-D (vectors are **bolded**)

	Position, $\boldsymbol{x}$ *or* $\boldsymbol{r}$	Velocity, $\boldsymbol{v}$	Acceleration, $\boldsymbol{a}$
Rectangular $(\boldsymbol{i},\boldsymbol{j})$	$x\boldsymbol{i}+y\boldsymbol{j}$	$\dot{x}\boldsymbol{i}+\dot{y}\boldsymbol{j}$	$\ddot{x}\boldsymbol{i}+\ddot{y}\boldsymbol{j}$
Path, normal-tangential $(\boldsymbol{e}_n,\boldsymbol{e}_t)$	N/A	$v\boldsymbol{e}_t$	$a_n\boldsymbol{e}_n+a_t\boldsymbol{e}_t$ $a_n=v\dot{\theta}=\frac{v^2}{\rho}=\rho\dot{\theta}^2$ $a_t=\dot{v}=\frac{dv}{dt}$
Polar $(\boldsymbol{e}_r,\boldsymbol{e}_\theta)$	$r\boldsymbol{e}_r$	$\dot{r}\boldsymbol{e}_r+r\dot{\theta}\boldsymbol{e}_\theta$	$(\ddot{r}-r\dot{\theta}^2)\boldsymbol{e}_r+(2\dot{r}\dot{\theta}+r\ddot{\theta})\boldsymbol{e}_\theta$

- Unit vectors for rectangular, path, and polar coordinate systems at a point on an arbitrary path



- Relative motion $\boldsymbol{v}_A=\boldsymbol{v}_B+\boldsymbol{v}_{A/B}$ (remember order!)
- Work and Energy
  - $T \stackrel{\text{def}}{=}$ kinetic energy; $T=\frac{1}{2}mv^2$
  - $V \stackrel{\text{def}}{=}$ potential energy (note that this is a capital *V* to distinguish between velocity)
  - $U=\int \boldsymbol{F}\cdot d\boldsymbol{r} \stackrel{\text{def}}{=}$ work (note that U_{NC} refers to work done by non-conservative forces)
- Impulse and Momentum
  - $\int \boldsymbol{F}dt \stackrel{\text{def}}{=}$ linear impulse
  - $m\boldsymbol{v} \stackrel{\text{def}}{=}$ linear momentum
  - $\int \boldsymbol{M}_P dt \stackrel{\text{def}}{=}$ angular impulse about a point P
  - $\boldsymbol{H}_P=\boldsymbol{r}_P\times m\boldsymbol{v} \stackrel{\text{def}}{=}$ angular momentum about point P, where $\boldsymbol{r}_P$ is the position vector from point P to the particle
- Coefficient of Restitution
  - $e=\frac{\int F_r dt}{\int F_d dt}=\frac{v_{B2}-v_{A2}}{v_{A1}-v_{B1}}$, where subscripts r and d refer to the restoration and deformation phases of a collision, respectively.

- Rigid Body Kinematics
  - $\omega \stackrel{\text{def}}{=}$ angular acceleration
  - $\alpha \stackrel{\text{def}}{=}$ angular acceleration
  - Velocity of arbitrary point A on a rigid body *with a fixed point*: $\boldsymbol{v}_A=\boldsymbol{\omega}\times\boldsymbol{r}_A$
  - Acceleration of arbitrary point A on a rigid body *with a fixed point*:
    - $\boldsymbol{a}_A=\boldsymbol{a}_{A,t}+\boldsymbol{a}_{A,n}=\boldsymbol{\alpha}\times\boldsymbol{r}_A+\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\boldsymbol{r}_A)=\boldsymbol{\alpha}\times\boldsymbol{r}_A-\omega^2\boldsymbol{r}_A$
      - (note that simplification of the last term is only valid in 2-D)
  - $\boldsymbol{v}_A=\boldsymbol{v}_B+\boldsymbol{v}_{A/B}$
  - $\boldsymbol{a}_A=\boldsymbol{a}_B+\boldsymbol{a}_{A/B}$

- Rotating Coordinate Systems
  - $\boldsymbol{v}_A = \boldsymbol{v}_B + \boldsymbol{v}_{A/B} = \boldsymbol{v}_B + \boldsymbol{\Omega}\, x\, \boldsymbol{r}_{A/B} + \left(\boldsymbol{v}_{A/B}\right)_{xyz}$
  - $\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B} = \boldsymbol{a}_B + \dot{\boldsymbol{\Omega}}\, x\, \boldsymbol{r}_{A/B} + \boldsymbol{\Omega}\, x\, (\boldsymbol{\Omega}\, x\, \boldsymbol{r}_{A/B}) + 2\boldsymbol{\Omega}\, x\, \left(\boldsymbol{v}_{A/B}\right)_{xyz} + \left(\boldsymbol{a}_{A/B}\right)_{xyz}$

- Mass moment of inertia: $I = \int r^2 dm$
  - Radius of gyration: $k = \sqrt{\frac{I}{m}}$
  - Parallel Axis Theorem: $I_A = I_G + md^2$, where the axes passing through points A and G are parallel, G refers to the center of mass, and d is the distance separating the two
- Equations of Motion (2-D)
  - $\boldsymbol{F} = m\boldsymbol{a}_G$
  - $\sum M_P = I_G\alpha \mp ma_G d$, where P is an arbitrary point, and d is the perpendicular distance between the line of action of ma_G and point P. The sign is determined by the direction of the moment produced by ma_G about P.
    - Special cases:
      - $\sum M_G = I_G\alpha$, where G is the center of mass
      - $\sum M_O = I_O\alpha$, where O is a fixed point
      - $\sum M_{IC} = I_{IC}\alpha$, where IC is the instant center of velocity for a circular cylinder that rolls without slip
- Work and Energy
  - $T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$
    - $= \frac{1}{2}I_O\omega^2$ (where O is a fixed point)
    - $= \frac{1}{2}I_{IC}\omega^2$ (where IC is the instantaneous center of velocity)
  - $U = \int M d\theta \stackrel{\text{def}}{=}$ work done by moment
- Angular Momentum
  - $H_1 + \int M dt = H_2$
  - $H_P = I_G\omega \mp mv_G d$ (P is an arbitrary point; d defined as above in moment equation)
    - Special cases:
      - $H_G = I_G\omega$, where G is the center of mass
      - $H_O = I_O\omega$, where O is a fixed point
      - $H_{IC} = I_{IC}\omega$, where IC is the instantaneous center of velocity
- Vibrations
  - $\omega_n \stackrel{\text{def}}{=}$ circular natural frequency (units are rad/s)
  - $f_n \stackrel{\text{def}}{=}$ natural frequency (units are Hz or cycles/s or revolutions/s)
  - $\omega_n = 2\pi f_n$
  - $\tau = 1/f_n \stackrel{\text{def}}{=}$ period of motion